# Paleomagnetism of the late Neoproterozoic Blueflower and Risky formations of the northern Cordillera, Canada[1]

**John K. Park**

**Abstract**: Paleomagnetic analysis of the Risky and underlying Blueflower formations at the top of the Neoproterozoic Windermere Supergroup in the northern Cordillera of Canada revealed several magnetic components of geological significance. Rock strata of the Blueflower locality in the Mackenzie Mountains yielded no primary remanence due to extensive "pyritization" and weathering. However, dolomitic rocks of the Risky locality in the eastern Wernecke Mountains, 250 km to the north, yielded a probable primary remanence. The remanence has a direction at $D$, $I$ = 055°, −64° ($N$ = 11 samples from 4 sites, $k$ = 18, $\alpha_{95}$ = 11°) and pole at 007°E, 29°S ($\delta p$, $\delta m$ = 14°, 18°), and indicates a paleolatitude of 46 ± 14°S. Comparison between poles from the Risky and the overlying Backbone Ranges and Ingta formations suggests a profound post-Risky unconformity. Three secondary components are also present. Component $B$ in both the Blueflower and Risky units appears to be a thermal overprint possibly related to Vendian rifting. Its respective direction in each unit is $D$, $I$ = 062°, +77° ($N$ = 4 sites, $k$ = 143, $\alpha_{95}$ = 8°) and $D$, $I$ = 064°, +56° ($N$ = 6 sites, $k$ = 21, $\alpha_{95}$ = 15°), yielding poles of 110°E, 64°S ($\delta p$, $\delta m$ = 13°, 14°) and 142°E, 44°S ($\delta p$, $\delta m$ = 15°, 21°). Component $C$, only found in the Risky, is attributed to Late Cambrian to Middle Ordovician extension and magmatism. Component $D$, which predominates in both formations, relates to late Mesozoic compression and to weathering.

**Résumé** : L'étude du paléomagnétisme de la Formation de Risky et de la Formation de Blueflower sous-jacente localisées au sommet du Supergroupe de Windermere, d'âge Néoprotérozoïque, dans la Cordillère septentrionale du Canada, met en évidence plusieurs composantes magnétiques géologiquement importantes. Les strates rocheuses de la localité de Blueflower dans les monts Mackenzie n'ont fourni aucune rémanence primaire due à la « pyritisation » étendue et à l'altération météorique. Cependant, les roches dolomitiques de la localité de Risky dans la région orientale des monts Wernecke, à 250 km au nord, ont fourni une rémanence primaire probable. La rémanence a une direction $D$, $I$ = 055°, −64° ($N$ = 11 échantillons de 4 sites, $k$ = 18, $\alpha_{95}$ = 11°) et un pôle à 007°E, 29°S ($\delta p$, $\delta m$ = 14°, 18°), indiquant une paléolatitude de 46 ± 14°S. La comparaison entre les pôles de la localité de Risky et des formations sus-jacentes de Backbone Ranges et d'Ingta suggère une discordance post-Risky profonde. Trois composantes secondaires ont aussi été décelées. Un composante $B$ dans les unités de Blueflower et de Risky est interprétée comme étant une surimpression thermique, probablement associée au développement d'un rift au Vendien. La direction respective de chacune des unités est $D$, $I$ = 062°, +77° ($N$ = 4 sites, $k$ = 143, $\alpha_{95}$ = 8°) et $D$, $I$ = 064°, +56° ($N$ = 6 sites, $k$ = 21, $\alpha_{95}$ = 15°), fournissant les pôles de 110°E, 64°S ($\delta p$, $\delta m$ = 13°, 14°) et 142°E, 44°S ($\delta p$, $\delta m$ = 15°, 21°). La composante $C$, observée uniquement dans le Risky, est attribuée à l'extension et au magnétisme durant l'intervalle Cambrien tardif − Ordovicien moyen. La composante $D$, laquelle prédomine dans les deux formations, est reliée à une phase de compression et à l'altération météorique durant le Mésozoïque tardif.
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## Introduction

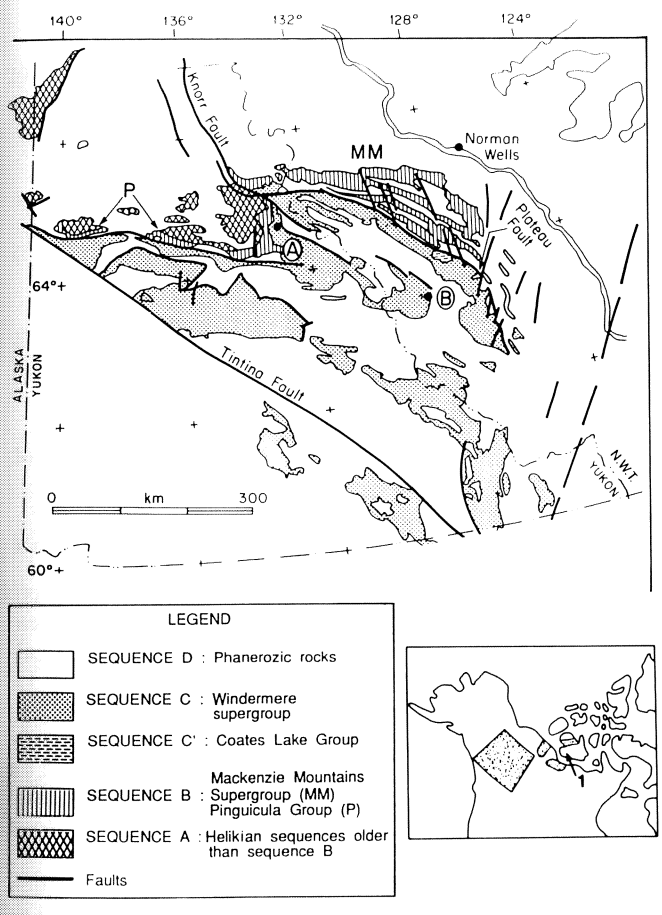
A recent interpretation of paleopoles from Laurentia places the "continent" over the south pole at some time during the late Neoproterozoic to Early Cambrian (Park 1992). In contrast, studies of lower Neoproterozoic and later lower Paleozoic strata indicate low paleolatitudes. The most reliable Neoproterozoic low-latitude poles come from Franklin igneous rocks (e.g., Fahrig et al. 1971) dated at 723 Ma (Heaman et al. 1992) and Mackenzie Mountains diabase (Park et al. 1989) dated at 779 Ma (Jefferson and Parrish 1989). It is unknown how Laurentia moved in the nearly 200 Ma period between 723 Ma and the Early Cambrian, because no reliable primary poles are currently available from this interval. Unfortunately, several poles from the Rapitan (Morris 1977)

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**Fig. 1.** Map showing sampled sections: locality A (circled), Risky Formation, eastern Wernecke Mountains (section 88AC-5, J.D. Aitken, unpublished data; 132°55′W, 64°39′N); locality B (circled), Blueflower Formation, Mackenzie Mountains (section 81AC-9, Aitken 1989; remeasured as section 88AC-2, J.D. Aitken, unpublished data; 128°30′W, 63°26′N). Inset shows general location map and location (1) of correlative B and C sequences (stippled areas) in Amundsen Basin of northern Canada. Sequence distribution is after Young et al. (1979) and Jefferson (1983). The Coates Lake Group (sequence C′) has been assigned to either the Mackenzie Mountains Supergroup (sequence B) (Jefferson and Parrish 1989; Park and Jefferson 1991) or the Windermere Supergroup (sequence C) (Aitken 1981; Gabrielse and Yorath 1989; Narbonne and Aitken 1992); in this study it is included with the Windermere.



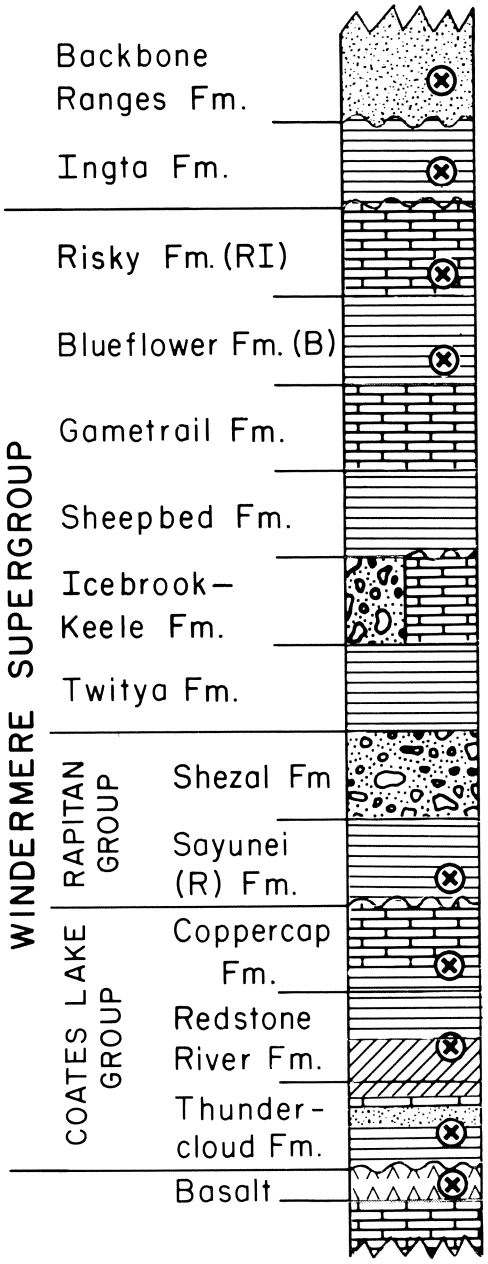

and Coates Lake groups (Park and Jefferson 1991) of the lower Windermere Supergroup, deposited during this period in the western Cordillera, may be affected by tectonic rotations (Park and Jefferson 1991; J.K. Park, unpublished data). The present study on strata at the top of the Windermere presents an attempt to better define the Neoproterozoic apparent polar wander (APW) path of Laurentia.

## Geology

### Setting and stratigraphy

The orogen and platform of the Cordillera of northwestern Canada contain two major Neoproterozoic sequences, represented in the Mackenzie Mountains by the older platformal

**Fig. 2.** Schematic stratigraphic column of the Windermere Supergroup. Lithic symbols are explained by reference to Backbone Ranges Formation (sandstone), Ingta Formation (siltstone), Risky Formation (carbonate), Shezal Formation (conglomerate), and Redstone River Formation (evaporite). Crosses indicate sampled units with corresponding studies as follows: Backbone Ranges – Ingta formations, Park (1992); Sayunei Formation, Morris (1977); Coates Lake Group, Park and Jefferson (1991); basalt, Morris and Aitken (1982).




Mackenzie Mountains Supergroup and the younger Windermere Supergroup (sequences B and C, Fig. 1). The Windermere Supergroup (Fig. 2), mainly a basinal succession, extends along the Cordillera into the United States. Locally developed rift strata of the Coates Lake and Rapitan groups of the basal Windermere are overlain by regionally developed passive-margin strata of the Twitya to Risky formations (Fig. 2). The Ingta Formation, atop the Risky in the Mac-

**Table 1.** *D* component from Blueflower Formation type section.

| Si | Level (m)[a] | Treatment AF (mT) | Treatment TH (°C) | *N*, *n* | *D* (°), *I* (°) In situ | *D* (°), *I* (°) Paleohor. | *R* | *k* | $a_{95}$ (°) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 175 | N– 30 | N– 150 | 6, 9 | 262, +74 | 058, +78 | 5.93 | 74 | 8 |
| | | 25– 90 | 150–>200 | 6, 6 | 269, +69[b] | 031, +81 | 5.87 | 39 | 11 |
| | 191 | 5– >20 | N– 225 | 5, 9 | 356, +85 | 076, +61 | 4.64 | 11 | 24 |
| | | — | 175–>300 | 3, 5 | 280, +74[b] | 070, +76 | 2.99 | 194 | 9 |
| 3 | 203 | 3– 15 | N– 250 | 5, 9 | 326, +69 | 052, +62 | 4.50 | 8 | 29 |
| 20 | 272 | N– 25 | N– 175 | 12, 14 | 065, +82 | 083, +51 | 11.69 | 35 | 7 |
| | | 3–>110 | 125– 400 | 12, 23 | 291, +78[b] | 075, +69 | 11.94 | 198 | 3 |
| 4 | 330 | 15– 45 | 250– 400 | 3, 6 | 154, +20[c] | 148, +09 | 2.89 | 19 | 29 |
| 5 | 342 | N– 15 | N– 300 | 6, 9 | 064, +84 | 079, +56 | 5.44 | 9 | 24 |
| 6 | 357 | N– 20 | N– 250 | 6, 13 | 306, +85 | 074, +68 | 5.71 | 17 | 17 |
| 7 | 364 | N– 45 | N– 300 | 6, 14 | 294, +82 | 075, +73 | 5.96 | 139 | 6 |
| 8 | 371 | N– 15 | N– 225 | 4, 6 | 039, +79 | 077, +52 | 3.70 | 10 | 31 |
| 9 | 405 | N– 10 | N– 200 | 3, 7 | 299, +80 | 060, +73 | 2.97 | 69 | 15 |
| 10 | 411 | N– 40 | N– 300 | 3, 6 | 292, +84 | 082, +70 | 2.99 | 200 | 7 |
| 11 | 460 | N– 25 | N– 150 | 6, 14 | 344, +68 | 046, +52 | 5.79 | 24 | 14 |
| 13 | 505 | N– 45 | N– 300 | 5, 9 | 285, +75 | 071, +73 | 4.98 | 185 | 6 |
| | | N– 35 | 175– 400 | 4, 5 | 282, +67[b] | 054, +79 | 3.95 | 60 | 12 |
| 12 | 515 | N– 15 | N– 250 | 2, 4 | 311, +78 | 065, +70 | 1.99 | – | – |
| Mean *D*, in situ, less 3, 5, 8: | | | | 10 | 309, +82 | — | 9.89 | 81 | 5 |
| Mean *D*, paleohorizontal: | | | | 10 | — | 069, +67 | 9.85 | 59 | 6 |
| Mean *B*, in situ (1, 2, 13, 20): | | | | 4 | 279, +72 | — | 3.99 | 207 | 6 |
| Mean *B*, paleohorizontal: | | | | 4 | — | 062, +77 | 3.98 | 143 | 8 |

**Notes:** Si, site number; AF, alternating field; TH, thermal; *N*, number of samples (or sites for grand means); *n*, number of specimens; *D*, *I*, declination and inclination of direction; *R*, vector resultant; *k*, Fisher's (1953) index of precision; $\alpha_{95}$, 95% confidence error about the mean direction. Selection criteria for means: for $N > 2$, $k > 10$; for $N = 2$, $R \geq 1.88$ (separation <40°).

[a]Stratigraphic height above base of Blueflower Formation (J.D. Aitken, unpublished data).

[b]Component *B*.

[c]Uninterpreted direction has normal and reverse polarities.

kenzie Mountains, was thought to represent the terminal Windermere (Aitken 1989), but has since been assigned, at least partly, to the Early Cambrian with the discovery of the fossil *Phycodes pedum* near the top (Narbonne and Aitken 1992).

A later rifting phase of Vendian age seems to follow the terminal Windermere. It is clearly defined in the southern Cordillera (Bond and Kominz 1984; Thompson et al. 1987), but in the northern Cordillera may be indicated by post-Risky graben structures into which later slump breccia and other sediments have been deposited (Fritz 1992). The Vendian rifting is separate from younger extensional tectonics of post-Early to pre-Late Cambrian age in the northern Cordillera that developed a system of basins and uplifts (Cecile 1982). Continued extension in the Middle Ordovician led to a major phase of alkalic volcanism (Cecile 1982). Folding and thrusting in the northern Cordillera occurred during late Mesozoic compression, possibly in two phases—a late Late Cretaceous phase (~100 Ma) and a Late Cretaceous–Paleogene (~65 Ma) phase (Eisbacher 1981). In the Mackenzie Mountains, the fold and thrust structures define an arc, convex to the northeast (Eisbacher 1981), that is primary in origin (Norris 1972; Aitken and Long 1978; Park et al. 1989).

## Blueflower Formation

Blueflower beds are predominantly composed of shale and mudstone, interspersed with subordinate turbidites (Aitken 1989). Mainly coarse sandstone samples (sites 1–13) were collected at the type section 81AC-9 (Aitken 1989; remeasured as 88AC-2, J.D. Aitken, unpublished data) at locality B (Fig. 1) (stratigraphic levels indicated in Table 1). Halfway up the section, samples were obtained from a 5 m thick debris-flow paraconglomerate bed (site 20), mainly consisting of limestone boulders. The section lies within an arcuate fold-fault belt characterized by steep longitudinal east- and west-dipping reverse faults (Blusson 1971). Rock strata at the collecting locality dip 24–35° to the east.

The Blueflower locality contains much pyrite, which occurs as single grains of cubic or framboidal aspect, and as overgrowths. Much of the pyrite has been altered by diagenesis and weathering, which at the type section has produced limonitic spots or a tan colouration due to the formation of goethite and probable lepidocrocite. Hematite is found as occasional blebs or patches of pigment in the limonite but rarely as defined grains (site 6). Magnetite is observed rarely at site 4, where it is partially altered to hematite. Rutile is sparse to common; it may replace other opaque minerals, including the secondary hematite staining.

## Risky Formation

The Risky Formation conformably overlies the Blueflower. At the collecting locality in upper Corn Creek of the eastern Wernecke Mountains (locality A (circled), Fig. 1), it is unconformably overlain by the Lower Cambrian Backbone

**Table 2.** Paleomagnetic directions from Risky Formation.

| Si | Level (m)[a] | Treatment AF (mT) | Treatment TH (°C) | N, n | D (°), I (°) In situ | D (°), I (°) Paleohor. | R | k | $\alpha_{95}$ (°) | Comp. |
|---|---|---|---|---|---|---|---|---|---|---|
| 25 | 504 | — | AF, 100– 640 | 3,[b] 3 | 003, +75 | 056, +74 | 2.97 | 32 | 22 | *D* |
| | | 20– 90 | — | 2, 4 | 035, +78 | 079, +70 | 1.98 | — | — | *B* |
| | | | 250– 560 | 3,[c] 4 | 240, +48 | 224, +54 | 2.92 | 24 | 26 | *A* |
| 32 | 295 | N– 20 | N– 540 | 3, 6 | 020, +39 | 034, +44 | 2.99 | 45 | 10 | *B* |
| | | — | 575– 665 | 7, 6 | 284, −03[d] | 284, +11 | 6.73 | 22 | 13 | *C* |
| 31 | 273 | 3– 25 | — | 4, 4 | 319, +75 | 071, +73 | 3.94 | 53 | 13 | $D_2$ |
| | | N– 200 | AF, 20– 100 | 4, 8 | 202, −83 | 277, −71 | 3.98 | 158 | 7 | $D_1$ |
| | | | | *8,*[b] *9* | *339, +81* | *092, +76* | *7.87* | *55* | *8* | *D* |
| | | — | 100–>450 | 5, 6 | 001, +56 | 031, +60 | 4.93 | 52 | 11 | *B* |
| | | 35– 55 | 400– 600 | 6, 9 | 266, −03[d] | 264, +14 | 5.58 | 12 | 20 | *C* |
| 30 | 267 | 10– 35 | 200– 500 | 2, 3 | 352, +78 | 076, +71 | 1.96 | — | — | $D_2$ |
| | | | | *3,*[b] *3* | *006, +79* | *080, +70* | *2.95* | *40* | *20* | *D* |
| | | | AF, 450– 500 | 2, 3 | 268, +15 | 264, +33 | 1.88 | — | — | *C* |
| 29 | 255 | — | N– 100 | 4, 10 | 345, +66[d] | 048, +76 | 3.97 | 104 | 9 | $D_1$ |
| | | | AF, 20– 100 | | | | | | | |
| | | — | 500– 660 | 2, 5 | 003, +75 | 086, +72 | 2.00 | — | — | $D_3$ |
| | | | AF, 200– 670 | | | | | | | |
| | | | | *6,*[b] *10* | *349, +69* | *062, +76* | *5.95* | *98* | *7* | *D* |
| | | | 450– 630 | 4 5 | 269, −05[d] | 268, +12 | 3.88 | 25 | 19 | *C* |
| 28 | 150 | 5– 25 | N– 400 | 5, 11 | 312, +76 | 090, +76 | 4.94 | 69 | 9 | $D_2$ |
| | | 20–>60 | 325– 500 | 3, 3 | 037, +71 | 079, +54 | 2.95 | 38 | 20 | *B* |
| | | | 500– 560 | 2, 3 | 101, −34 | 094, −59 | 1.99 | — | — | *A* |
| 27 | 144 | N– 28 | 100– 520 | 5, 11 | 349, +70 | 098, +64 | 4.98 | 231 | 5 | $D_2$ |
| | | 30– 110 | 375– 530 | 5, 8 | 025, +52 | 071, +46 | 4.82 | 19 | 18 | *B* |
| | | | 500– 640 | 4, 4 | 085, +22 | 089, −06 | 3.90 | 31 | 17 | *C* |
| | | | 520– 575 | 2, 2 | 086, −33 | 052, −54 | 1.91 | — | — | *A* |
| 26 | 143 | N– 35 | AF, 100– 400 | 4, 5 | 351, +71 | 104, +59 | 3.95 | 57 | 12 | $D_2$ |
| | | N– 110 | N– 520 | 5, 8 | 023, +64 | 090, +48 | 4.90 | 40 | 12 | *B* |
| | | | 500– 560 | 4, 4 | 109, −45 | 039, −76 | 3.94 | 48 | 13 | *A* |
| Mean $D_2$, | in situ: | | | 5 | 338, +74 | — | 5.92 | 169 | 6 | |
| | paleohor.: | | | 5 | — | 090, +69 | 5.86 | 96 | 8 | |
| Mean *D*, | in situ: | | | 7[e] | 347, +75 | — | 6.97 | 178 | 5 | |
| | paleohor.: | | | 7[e] | — | 088, +72 | 6.92 | 76 | 7 | |
| Mean *C*, | in situ: | | | 5 | 270, −04 | — | 4.86 | 29 | 15 | |
| | paleohor.: | | | 5 | — | 270, +15 | 4.90 | 39 | 12 | |
| Mean *B*, | in situ: | | | 6 | 021, +60 | — | 5.82 | 29 | 13 | |
| | paleohor.: | | | 6 | — | 064, +56 | 5.77 | 21 | 15 | |
| Mean *A*, | in situ: | | | 4 | 090, −41 | — | 3.86 | 22 | 20 | |
| | paleohor.: | | | 4 | — | 059, −62 | 3.89 | 27 | 18 | |

**Notes:** Symbols defined in Table 1. AF indicates that specimens were demagnetized in alternating fields from 50–180 mT before thermal treatment. Italicized row entries indicate means of *D* subcomponents.

[a]See Table 1.

[b]Each *D* subcomponent in a sample is treated as an independent determination (i.e., treated as an extra sample) from the site.

[c]Outlier direction, 79° from mean, not used.

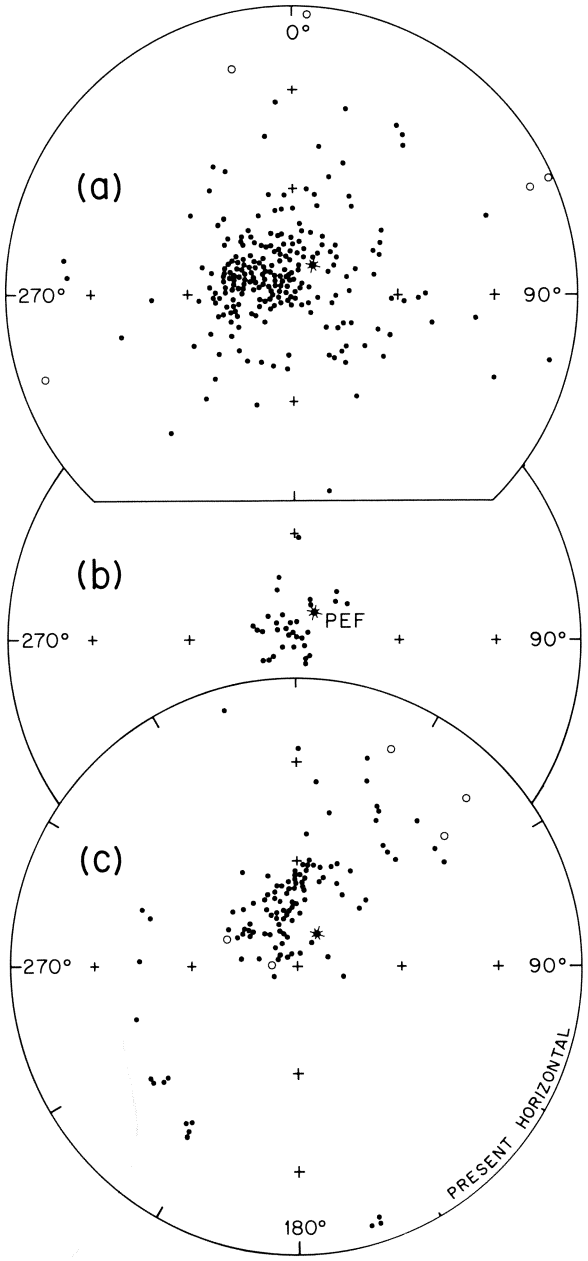
[d]Normal and reverse directions.

[e]Excluding site 32.

Ranges Formation. Risky strata (section 88AC-5, J.D. Aitken, unpublished data) consist of dolomites and dolomitic-quartz sandstones at the top and base of the section and mainly siltstones and quartz sandstones in between. Beds dip 14–43° to the southeast. Four sites (25–28) were collected from dolomitic-quartz sandstones and four (29–32) from fine- to medium-grained quartz sandstones in a predominantly silty part of the section (stratigraphic levels are given in Table 2). The locality occurs on an uplift, the Ogilvie Arch, caused by Cambrian extension, which also produced the Misty Creek embayment, a long-lived basin 50–60 km to the east (Cecile 1982).

The Risky locality appears to be less weathered than the Blueflower. Weathering is more in evidence at the quartzitic sites, which contain limonite, including goethite, derived from pyrite of which there are scarce remnants. Limonite is sparsely to abundantly developed and fills fractures. Dolomitic sites (25–28) contain some hematite but mainly altered opaque grains (<30 μm), including rutile; quartzitic sites (29–32) contain larger opaque grains (<200 μm) altered to

**Fig. 3.** Natural remanent magnetizations of specimens from sandstones (*a*) and conglomerate (*b*) of the Blueflower Formation and from the Risky Formation (*c*). Closed circles (open circles) represent downward (upward) directed magnetizations. The star represents the direction of the present Earth's field (PEF) at the sampling localities.




rutile. Few sulphide grains are present. Use of a scanning electron microscope (SEM) identified iron sulphides, iron oxides, and iron–titanium oxides, up to 1 $\mu$m in size, in all sites.

## Methods

In the field, from two to seven hand samples, oriented with sun and magnetic compasses, were collected at each site except the conglomerate, where cores were drilled from 13 boulders. In the laboratory, standard cylindrical specimens (2.2 cm long, 2.5 cm in diameter) were prepared for study.

Magnetic measurements were made with a Geofyzika JR-4 fast spinner magnetometer. After measurement of their natural remanent magnetizations (NRM's), specimens were subjected to alternating field (AF) and (or) thermal (TH) treatment to resolve the NRM. Instrumentation and procedures are discussed in Park and Jefferson (1991). Demagnetization was generally terminated when magnetic intensities became too weak to measure, samples acquired a significant spurious component, or AF's proved ineffective in removing the magnetization. In the second case, spurious components include the acquisition during AF treatment of gyroremanent magnetization, characterized by "zig-zag" behaviour in vector diagrams (Park 1984), and the acquisition during thermal treatment of magnetization due to chemical changes. In the last case, demagnetization was continued in a magnetically shielded room. Remanence data were analyzed using vector diagrams and the LINEFIND (Kent et al. 1983) and Linear spectrum analysis (Schmidt 1982) programs. The latter program evaluates data from a number of specimens and identifies common components.

Anisotropy of magnetic susceptibility (AMS) of Risky specimens, both untreated and thermally treated, was carried out using a Geofyzika Kappabridge KLY-2 magnetic susceptibility meter. All susceptibility experiments were done after demagnetization experiments were complete.

## Results and analysis

### Paleomagnetic components

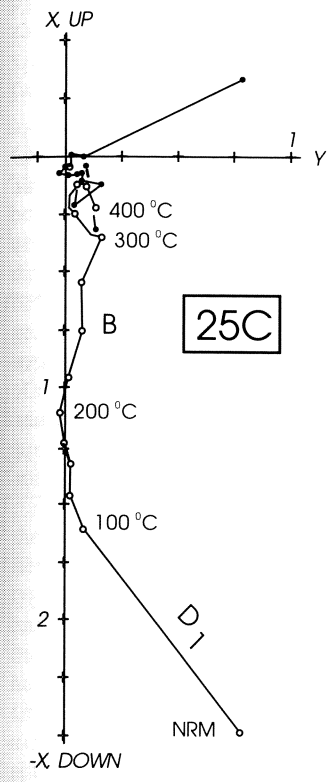
*Blueflower Formation*

Most in situ NRM directions are inclined steeply down to the northwest, away from the present Earth's field (PEF) direction in the collecting region (Figs. 3*a*, 3*b*). AF and (or) TH demagnetization yielded at least three magnetic components. The predominant component, *D*, has laboratory unblocking temperatures ($T_{UB}$'s) up to 300°C (Figs. 4, 5) and coercivities to 45 mT. Its direction lies near the PEF direction (*D*, $I = 032°, +79°$) in the region (Table 1). It likely resides in the observed goethite and other oxyhydroxides, the products of pyrite alteration. The second component, *B*, also with a steeply inclined downward direction, has distributed $T_{UB}$'s to 400°C (Figs. 4, 5) and coercivities that may exceed 110 mT. In the conglomerate site 20, the coherent direction of *B* from all the boulders shows it to be secondary (Fig. 5; Table 1). A third component with higher $T_{UB}$'s than *D* or *B*, was only found at site 4, which lacks a stable *B* component, and at site 20 in one boulder of the conglomerate (Fig. 5*b*). It may reside in the observed magnetite at site 4. Site directions and means of *B* and *D* are plotted in Fig. 6 and given in Table 1.

*Risky Formation*

NRM directions of the Risky Formation are similar to those of the Blueflower (Fig. 3*c*) and, similarly, contain a predominant *D* component (Table 2). *D* is probably carried by two or three different magnetic phases, judging by $T_{UB}$'s and coercivities. $T_{UB}$'s below 100°C combined with high coercivities to 200 mT (Fig. 7*a*), and $T_{UB}$'s to 670°C characterize subcomponents $D_1$ and $D_3$, respectively, which probably reside in goethite and derivative hematite formed from recent weathering of pyrite. $D_1$ has normal or reverse polarity. The third subcomponent, $D_2$, has intermediate $T_{UB}$'s to 550°C

**Fig. 4.** Vector diagram of a thermally treated sandstone specimen from site 2 of the Blueflower Formation plotted in the horizontal ($X-Y$) and vertical ($UP-Y$) planes showing approximate locations of components $B$ and $D_1$. Closed (open) circles are on the horizontal (vertical) plot. Intensities are in mA/m.



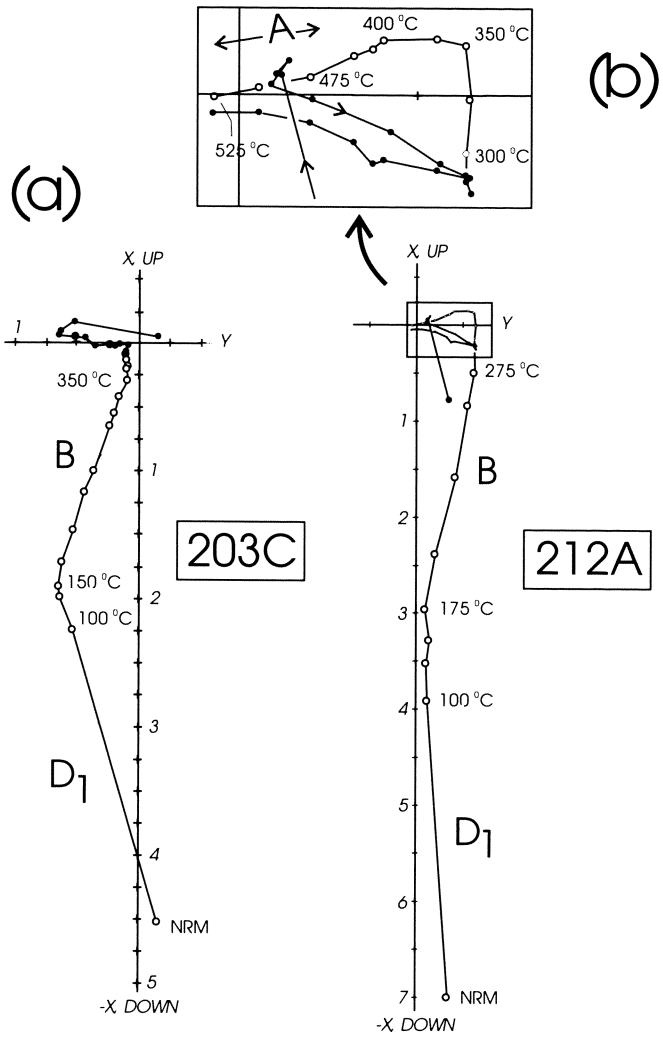

and coercivities to 35 mT (Figs. 7*a*, 8*a*), and is revealed after AF treatment by TH treatment to about 550°C. The mineral carrying $D_2$ is unknown.

The $T_{UB}$ ranges of $D_2$ overlap those of the $B$ component resolved in the $T_{UB}$ range to 540°C and the coercivity range to 110 mT (Fig. 8*a*) (Table 2). Unlike $D_2$, however, $B$ is not revealed by TH treatment after AF demagnetization. Where both $D_2$ and $B$ occur in the same specimen, $B$ is revealed in the higher $T_{UB}$ and coercivity ranges (Fig. 8*a*).

Two other components, $A$ and $C$, were also revealed: $A$ occurs exclusively in the "dolomitic" rocks (sites 25–28) and $C$, mainly in the quartzitic rocks (sites 29–32). $A$ has moderate to steep inclinations and "magnetite" $T_{UB}$'s (250–575°C) (Fig. 8*b*), while $C$ has shallow inclinations and either "magnetite" (200–560°C) or "hematite" $T_{UB}$'s (550–665°C) (Figs. 7*b*, 8*b*) (Table 2). $C$ is rarely resolved by AF treatment, and $A$, never. Both show reversals of polarity; however, reversals of $C$ commonly occur within single specimens (Figs. 7*b*, 8*b*?) or sites, whereas reversals of $A$ occur between different sites. The "magnetite" and "hematite" phases carrying $C$ may record reverse or normal single-polarity remanences, or dual-polarity remanences, with magnetite harbouring the reverse direction and hematite, the normal. In one instance, a single specimen records a reversal of $C$ in the hematite $T_{UB}$ range.

$A$ and $B$ seem to reside in magnetite, because, besides the magnetic characteristics, neither is resolved by TH treatment after AF demagnetization to 100 mT. In addition, $B$ seems

**Fig. 5.** Vector diagram of thermally treated specimens from the conglomerate site 20 sampled in the Blueflower Formation, showing components $A$, $B$, and $D_1$. See Fig. 4 for other explanations.




to be a thermal overprint on the assemblage of grains carrying $A$, because, where $B$ and $A$ (and occasionally $D_2$) occur in the same specimen, $A$ has higher $T_{UB}$'s to 575°C that appear complementary to $B$ (Fig. 8*b*). Site mean directions of $A$, $B$, $C$, and $D$ are plotted in Fig. 6.

Fold tests (McFadden 1990) were applied to all components of the Risky and Blueflower formations. Tests on the Blueflower $D$ and Risky $A$ and $C$ components gave equivocal results. The test on the $A$ component gave a higher test statistic ($\zeta_2 = 2.027$) after unfolding but below the 95% critical value (2.33); however, only four sites carry the remanence. On the other hand, a test on the Risky $D$ component indicates either a prefolding (test statistic $\zeta_2 = 4.785$ in situ and 3.930 after tilt correction for $N = 7$; 99% critical value = 4.253) or synfolding magnetization. A minimum $\zeta$ occurs at 33% unfolding (McFadden 1990) and a maximum $k$ value (precision) occurs at 14% unfolding (McElhinny 1964), supporting the synfolding possibility. A fold test on the $B$ component was equivocal for both formations, but proved significant at the 99% level, when sites from both units were combined (test statistic $\zeta_2 = 6.215$ in situ and 1.701 after

**Fig. 6.** Site directions and means from the Blueflower and Risky formations. Reversed polarity *A* and *C* directions are circled. *A*, *B*, and *C* are plotted with respect to the paleohorizontal; *D*, with respect to the present horizontal. In the inset, *D* directions are compared to path of directions calculated from poles given in Irving and Irving (1982) for the period 140–10 Ma.

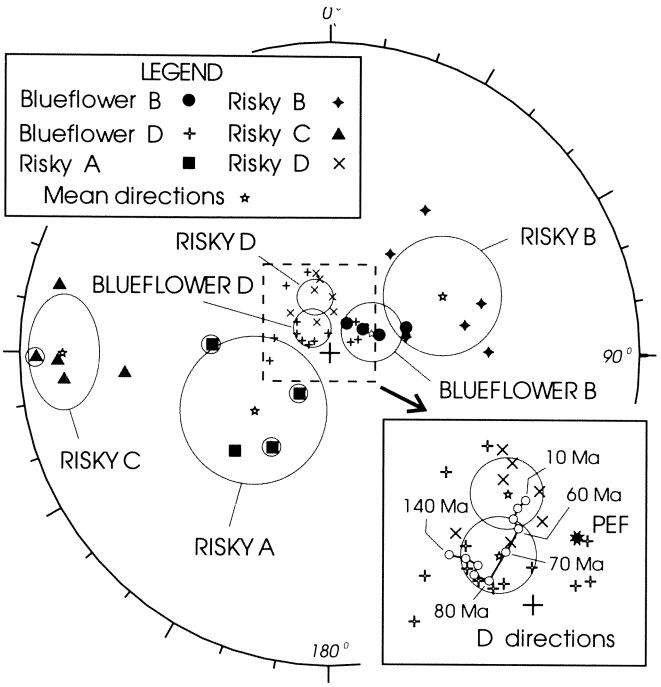
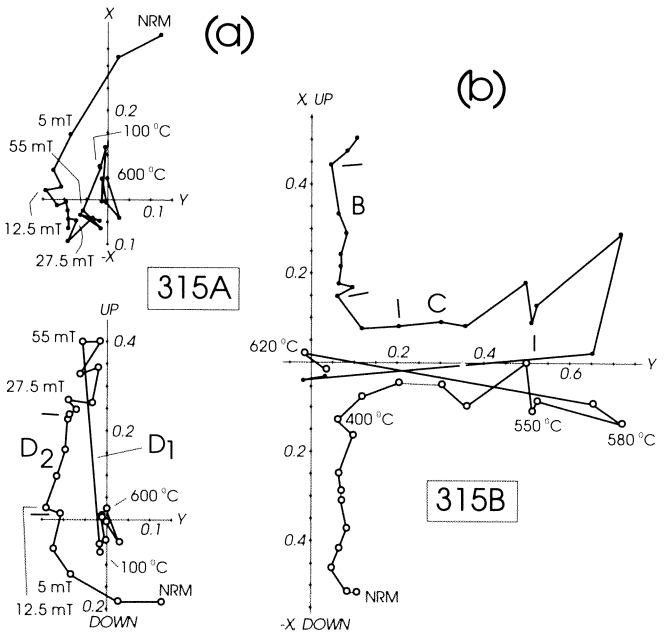
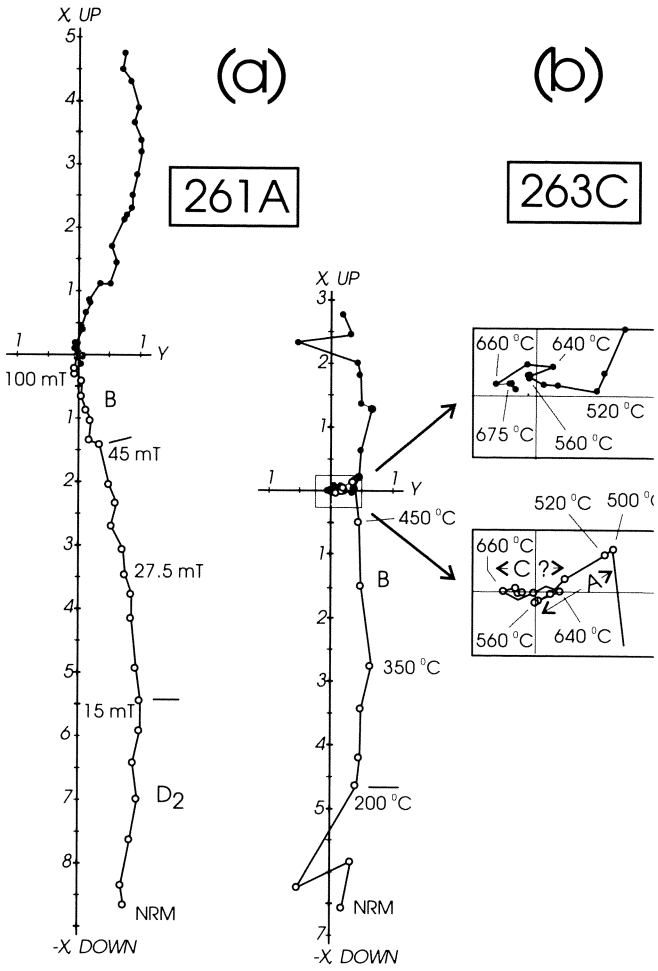



**Fig. 7.** Vector diagrams of two thermally treated specimens from a single sample of the quartz sandstone from site 31 showing approximate positions of *B*, *C*, and *D* components. See Fig. 4 for other details.




tilt correction for $N = 10$; 99% critical value = 5.120). It suggests that *B* predates folding. A minimum $\zeta$ occurs at 33% unfolding (McFadden 1990); however, as the maximum $k$ occurs at 100% unfolding (McElhinny 1964) and the mean

**Fig. 8.** Vector diagrams of AF (*a*) and thermally treated (*b*) specimens from dolomitic-quartz sandstone site 26 of the Risky Formation showing approximate positions of components *A*, *B*, and $D_2$. Other explanations are given in Fig. 4.




direction ($D$, $I$ = 064°, +65°) is >25° away from directions acquired at about the time of the Cretaceous–Paleogene compression (Fig. 6), *B* is thought to considerably predate the folding.

## Discussion

### Problems with analysis

Recovery of interpreted primary magnetization and the analysis of NRM is rendered difficult by the generally weak remanence intensities, the development of an occasional gyroremanent magnetization during AF treatment, the presence of weathering, and chemical alteration during TH treatment.

*Diagenesis and weathering*

Presence of extensive pyrite, lack of magnetic iron minerals, and replacement of other opaque minerals by rutile in the Blueflower Formation suggest that primary ilmenite, titanomagnetite, and titanohematite grains have been converted to rutile during "pyritization" (Ramdohr 1969). Pyritization of the Risky Formation appears to be less intense; pyrite is virtually absent at the dolomitic sites. Recent alteration of pyrite has led to the development of limonite in the Blueflower and

Risky formations. Van der Voo and Lowrie (1979) found that blue-gray limestones, containing primary magnetite and pyrite, could be used for determining primary paleomagnetic directions, but that beige- or tan-coloured varieties, where pyrite had been converted to goethite, could not be used. These observations are borne out in the present study of the pyritized Blueflower Formation, where a primary magnetization was not resolved. The dolomitic Risky sites, which yielded a possible primary component (discussed below), range from gray to gray with a slight orange or pink tinge.

*Chemical alteration*

Thermal treatment produced obvious chemical alterations in both the Blueflower and Risky formations. Growth of magnetic phases in Blueflower samples are generally indicated by directional instability and magnetic intensity increases between 200 and 400°C. Although the Risky Formation is generally much more stable to chemical alteration, AMS measurements indicated changes in bulk susceptibility and other AMS parameters between unheated samples and those heated to 600°C or higher. On heating, the most dramatic susceptibility increases occur in the quartzitic sites, which all contain limonite.

Chemical changes due to TH treatment did not produce the analyzed components. In the Risky Formation, component *C* is found in both the quartzitic and dolomitic rock types, is occasionally revealed by AF demagnetization, and has dual polarities in some specimens: these facts together indicate that it does not result from heating. Component *A* has a discreet $T_{UB}$ range, which may be followed by that of *C*: this likely indicates that it is not a reduction product from a hematite precursor. Measurements obviously affected by spurious magnetizations were not used.

## Magnetizations and their poles

Extensive pyritization and weathering, combined with the effects of chemical alteration during thermal experiments, have served to remove and (or) mask a primary component in the Blueflower Formation. The shallowly inclined component at site 4 is likely secondary. Components *D* and *B* in both formations are also secondary. In the Risky and Blueflower formations, the fold test, combined with the directional agreement of *D* components with a reference path traced by Cretaceous and later directions (calculated from poles of Irving and Irving (1982); Fig. 6), indicates that *D* is partly due to synfolding diagenetic processes. The presence of *D* subcomponents in the Risky Formation, along with the evidence of weathering in both units, suggests that *D* is also partly due to Recent weathering. These magnetizations due to later diagenesis and weathering cannot be reliably separated.

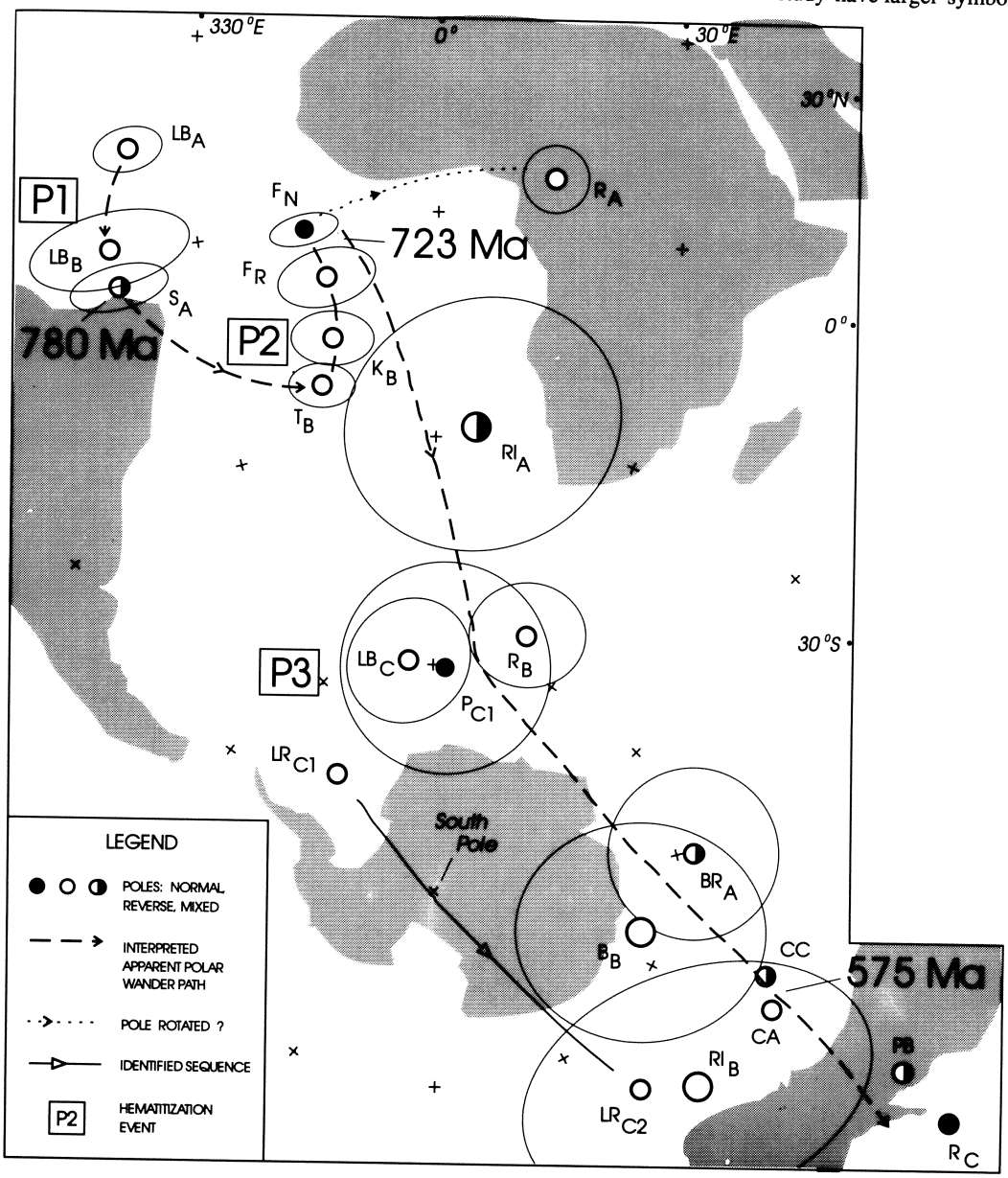
The *B* directions of the Blueflower and Risky formations, though distinct from *D*, also lie near the PEF direction in the region (Fig. 6). A fold test on each unit separately, assuming the *B* components to be unrelated, does not rule out a recent origin. However, a fold test involving the two units, combined with the difference between the *B* directions of the Blueflower and especially Risky and the Cretaceous–Paleogene reference path (Fig. 6), suggests, rather, that they considerably predate the folding. *B* poles from the Blueflower ($B_B$) and Risky ($RI_B$) formations plot near a combined primary pole from the overlying Ingta and Backbone Ranges formations ($BR_A$, Park 1992) (Fig. 9; Table 3) and poles (CC, Symons and Chiasson 1991; CA, Meert et al. 1994) dated at about 575 Ma. This suggests that they could relate to the Vendian period. The conglomerate test indicates that *B* is secondary, and two possibilities are offered for its origin. First, *B* could relate to extensive erosion, weathering, and alteration at the top of the Risky Formation prior to Cambrian sedimentation (Osborne et al. 1986). This possibility is discounted because sites are located between 25 and 385 m below the level of Vendian erosion, which has beveled off the ancient weathered part at the Risky locality. Second, *B* could relate to a thermal overprint caused by either burial or rifting in Vendian time. Burial heating is ruled out because a lack of evidence for compaction (Fred Chandler, personal communication, 1994) suggests that Risky strata were never deeply buried. On the other hand, rifting, as probably documented by the 575 Ma poles (CA, CC; Fig. 9) (Meert et al. 1994) of Laurentia, is also documented for Vendian to Early Cambrian times from other parts of the world (Bond et al. 1984).

Component *C* is difficult to interpret. It probably comprises secondary chemical magnetizations in magnetite and hematite, because it occurs with mixed polarities within sites (Table 2) and, according to evidence presented below, it is not primary. This explanation is consistent with the fact that *C* is predominantly found in the coarser, more porous quartzitic sites, which may have been more susceptible to chemical alteration by fluids. There is no positive fold test on *C*, but its shallowly inclined direction clearly indicates that it predates Mesozoic folding, which is associated with steeper inclined directions. A possible explanation is that the magnetic phases carrying *C* developed from the more common pyrite (Suk et al. 1990) in the quartzitic sites. Its pole ($RI_C$; 07°N, 140°E) (Table 3) is similar to Late Cambrian (mean of 10°N, 159°E; $A_{95}$ = 11°) (Symons and Chiasson 1991) to Middle Ordovician poles, consistent with a post-*B* age. In fact, *C* may be related to post-Early Cambrian to Middle Ordovician extension and magmatism that affected a large region of the northern Cordillera (Cecile 1982).

Component *A* comes from only four sites, has a large error, and usually has a $T_{UB}$ range between that of *B* and *C*. Together, these facts suggest that *A* could be an artifact of the data, namely, a vector resultant of *B* and *C*. Moreover, in the only site (27) where *A*, *B*, and *C* are revealed consistently together, *A* has a large error, which one might consider to result from vectorial combinations of *B* and *C* in differing proportion. However, if one adds the *B* and *C* vectors from this site, the resultants are not directed upward with moderate to steep inclinations characteristic of *A* but, rather, downward with steep inclinations (e.g., site 26, Fig. 8*b*). In addition, there is the evidence that *A* and *B* have complementary magnetic spectra. *A* is therefore considered to be a bonafide component.

Component *A* is only found in the dolomitic sites as a distinct single-polarity remanence (Fig. 6), either normal or reverse. Its characteristic $T_{UB}$'s indicate that it may reside in submicroscopic magnetite or titanomagnetite, corresponding to the generic iron and iron–titanium oxides identified in SEM studies. The fact that *A* is only found in the fine-grained and well-cemented dolomitic sites, and not in the more easily weathered quartzitic sites that also host probable younger components *C* and *D*, suggests that it predates these components. Because there is a reverse magnetization at one site,

**Fig. 9.** Apparent polar wander path of Laurentia for the Neoproterozoic and Early Cambrian after Park (1992, 1994). Paleopoles in alphabetical order are as follows: $B_B$, Blueflower Formation (this study); $BR_A$, Backbone Ranges – Ingta formations (Park 1992); CA, Catoctin Formation (Meert et al. 1994); CC, Callander complex, Ontario (Symons and Chiasson 1991); $F_N$ and $F_R$, normal and reverse Franklin diabases, Canadian Shield (references in Park and Jefferson 1991); $K_B$, Katherine Group (Park and Aitken 1986*a*); $LB_A$, $LB_B$, and $LB_C$, basinal sequence of the Little Dal Group (Park 1981*a*); $LR_{C1}$ and $LR_{C2}$, Rusty Shale formation of the Little Dal Group (Park and Jefferson 1991); $P_{C1}$, secondary component from Escape Rapids and Nelson Head formations of the Brock Inlier, N.W.T. (Park 1991); PB, Puerto Blanco volcanoclastic rocks (Barr and Kirschvink 1983); $R_A$, $R_B$, and $R_C$ ($R_X$, $R_Y$, and $R_Z$ of Morris 1977); $RI_A$ and $RI_B$, Risky Formation (this study); $S_A$, diabase in the northern Cordillera (Park 1981*b*; Park et al. 1989); $T_B$, Tsezotene Formation (Park and Aitken 1986*b*). P1–P3 refer to periods of secondary remanence acquisition. The error about poles is $\delta p/\delta m$, calculated from mean paleomagnetic directions; errors about $R_A$ and $R_B$ are derived from specimen averages; errors about $P_{C1}$ and $RI_A$ are from sample averages; errors about $LR_{C1}$ and $LR_{C2}$ are $\delta p/\delta m = 25°/29°$ and 27°/35°, respectively. Poles derived from this study have larger symbols.




*A* could be primary component; however, there are no field tests to verify its nature.

Could *A* be secondary? *A* is not easily explained as a Phanerozoic component from arguments given above. In addition, its absence from the more easily weathered quartzitic sites suggests that it was not acquired during diagenesis in the Vendian hiatus following Risky sedimentation. The P3 overprint poles (Fig. 9), instead, may represent that dia-

**Table 3.** Poles[a] and paleolatitudes.

| Symbol | Reference plane | Location (°W, °N) | Sites, Sa | Pole (°E, °S) | $\delta p$, $\delta m$ (°) | Polarity | $\lambda_P$ (°S) |
|---|---|---|---|---|---|---|---|
| $B_D$ | Present horizontal | 128.5, 63.4 | 10, 54 | 193, −70 | 9, 11 | N | −74±9 |
| $B_B$ | Present horizontal | 128.5, 63.4 | 4, 25 | 171, −52 | 10, 11 | N | −57±11 |
| | Paleohorizontal | 128.5, 63.4 | 4, 25 | 110, 64 | 13, 14 | R | 65±13 |
| $RI_D$ | Present horizontal | 132.9, 64.6 | 7, 26 | 122, −84 | 8, 8 | M | −62±8 |
| $RI_B$ | Present horizontal | 132.9, 64.6 | 6, 23 | 010, −64 | 15, 19 | N | −41±15 |
| | Paleohorizontal | 132.9, 64.6 | 6, 23 | 142, 44 | 15, 21 | R | 36±15 |
| $RI_C$ | Present horizontal | 132.9, 64.6 | 5, 23 | 136, 01 | 7, 15 | M | 2±7 |
| | Paleohorizontal | 132.9, 64.6 | 5, 23 | 140, −07 | 7, 13 | M | −8±7 |
| $RI_A$ | Present horizontal | 132.9, 64.6 | 4, 11 | 148, −21 | 15, 24 | M | −24±15 |
| | Paleohorizontal | 132.9, 64.6 | 4, 11 | 003, 28 | 22, 28 | M | 44±22 |
| | Paleohorizontal | 132.9, 64.6 | 11 | 007, 29 | 14, 18 | M | 46±14 |

**Notes:** $B_D$ and $B_B$ are Blueflower Formation *D* and *B* components; $RI_D$, $RI_C$, $RI_B$, and $RI_A$ are Risky Formation *D*, *C*, *B*, and *A* components. Sites, number of sights; Sa, number of samples. $\delta p$, $\delta m$, semiaxes of the oval of 95% confidence about the pole calculated from the mean site directions; $\lambda_P$, paleolatitude of the sampling locality.

[a]Calculation of poles is based on a postulate of this paper that Laurentia was in the southern hemisphere during the Neoproterozoic Era.

genetic event (Park 1994); they have a significantly different direction from $RI_A$, the pole of *A*, and are derived from components in hematite or related phases. $RI_A$ (Table 3) lies between older poles of Windermere age ($T_B$, $K_B$, $F_R$, $F_N$; see Park and Jefferson 1991) and the younger P3 poles, and plots on a proposed path (Park 1992) that ends with still younger poles of Vendian or Early Cambrian age ($BR_A$, CC, CA). Because $RI_A$ lies in stratigraphic sequence and does not agree in direction with any documented secondary poles in the region, it is considered to be primary. Its direction indicates a mid-paleolatitude position for the northern Cordillera. The 60° difference in position between the pole $RI_A$ and the pole $BR_A$ from immediately overlying strata is interpreted to indicate that the post-Windermere hiatus represents a profound unconformity.

There is no present test to indicate whether the position of $RI_A$ has been shifted because of tectonic rotation. Local rotations have been identified by paleomagnetism in the Coates Lake (Park and Jefferson 1991) and Rapitan groups (J.K. Park, unpublished data) west of the Plateau Fault, 250 km to the east (C′, Fig. 1). There is also suggestion that the Wernecke area was translated northward along "a series of dextral transcurrent faults and in part rotated about 30° counterclockwise" sometime during Rapitan time (Bell 1982); but this event would predate Risky deposition. The Risky locality is outside the region of Cambrian extension to the east (Cecile 1982), and, according to Norris and Hopkins (1977), "there is no evidence for large post-Early Paleozoic strike-slip movements" along the Richardson Fault Array, represented by one fault northeast of the locality that blends into Knorr Fault (Fig. 1). The location of $RI_A$ by Neoproterozoic poles from the Mackenzie Mountains suggests that it has not been affected by appreciable rotation (Fig. 9).

## Conclusions

1. The principal component, *D*, steeply directed in both the Risky and Blueflower formations, probably relates to diagenesis during late Mesozoic folding and to weathering.

2. Component *B*, also steeply directed in both the Risky and Blueflower formations, is interpreted to be a thermal overprint due to Vendian rifting.

3. Component *C*, a shallowly inclined component in Risky sites, is attributed to a secondary overprint developed in Late Cambrian to Middle Ordovician time during extension.

4. Extensive pyritization and weathering, combined with chemical alterations during heating, precluded finding a primary magnetization in the Blueflower Formation.

5. Component *A*, from fine-grained dolomitic rocks of the Risky Formation, is interpreted as a primary magnetization and indicates a mid-paleolatitude for Laurentia.

6. The directional difference between the interpreted primary poles from the Risky and the overlying Backbone Ranges – Ingta formations suggests that the post-Windermere hiatus represents a profound unconformity.

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